

Explaining Legal Equation Transformations

Answer Key

Algebra 1

Quick Answers

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|---|----------------------------------|
| 1. 5, 22 | 6. 13 |
| 2. 32 | 7. 0, 4 |
| 3. $2x + 10$ | 8. 4 |
| 4. 3, 12 | 9. $-\infty$, -1, <i>loopen</i> |
| 5. -4, ∞ , <i>built-in function open</i> | 10. 7 |
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Curriculum

Level: Algebra 1

HSA-REI.A.1 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–4 of 5 across 12 tagged checks.

Common wrong answers

- If they got 8: undid the multiplication before the addition, dividing only the 22 by 3*
 - If they got 6: subtracted 3 from the right side instead of adding 3 to both sides, which gives 9/5*
 - If they got 5: substituted into the changed equation instead of the original one*
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How to use this key. A step is legal when the new equation has the same solution set as the old one. Credit the *reason*, not just the arithmetic: a student who reaches the right number while writing “subtracted 7” on a line where 7 was subtracted from one side only has not yet learned this skill.

- 1.** Solve $3x + 7 = 22$ with reasons, then check.

Solution: From $3x + 7 = 22$, subtract 7 from *both* sides — Subtraction Property of Equality — to get $3x = 15$. Then divide both sides by 3, which is legal because 3 is not zero — Division Property of Equality. That gives $\boxed{x = 5}$.

Check: substituting into the original left side, $3(5) + 7 = 15 + 7 = \boxed{22}$, which matches the right side, so the solution set really was preserved. A student who divides first, turning 22 into $\frac{22}{3}$ and then subtracting 7, lands at $x = \frac{22}{3} - 7$; putting that back gives 8, not 22, which is exactly how the check catches an illegal order.

- 2.** Solve $\frac{x}{4} - 3 = 5$ with reasons.

Solution: Add 3 to both sides — Addition Property of Equality — giving $\frac{x}{4} = 8$. Multiply both sides by 4 — Multiplication Property of Equality — giving $\boxed{x = 32}$. The addition comes first because the subtraction is the operation applied *last* to x on the left, and legal steps undo operations in reverse order.

3. Rewrite $2(x + 5)$ and name the property.

Solution: The Distributive Property gives $2(x + 5) = 2 \cdot x + 2 \cdot 5$, that is $\boxed{2x + 10}$. This one is different in kind from the others: nothing was done to *both* sides. Only one side was rewritten into a form equal to it for every value of x , so no equation containing it can gain or lose a solution. That is the second family of legal transformations — rewriting a side using an identity.

Kind of step	Why it is legal
Same operation on both sides	Both sides stay equal, so solutions are kept
Rewrite one side by an identity	That side's value is unchanged for every x

4. Devi: $5x - 3 = 12 \rightarrow 5x = 9 \rightarrow x = \frac{9}{5}$.

Solution: The second line is the illegal one. Devi subtracted 3 from the right side only ($12 - 3 = 9$) instead of adding 3 to both sides. Nothing was done to the left side at all, so the two sides were no longer changed equally and the solution set moved.

Legal version: add 3 to both sides — Addition Property of Equality — giving $5x = 15$; then divide both sides by 5 — Division Property of Equality — giving $\boxed{x = 3}$.

Check: $5(3) - 3 = 15 - 3 = \boxed{12}$, matching the right side. Devi's value fails the same check: $5 \cdot \frac{9}{5} - 3 = 6$, not 12.

5. Solve $-2x < 8$.

Solution: Divide both sides by -2 . Dividing an inequality by a negative number reverses the direction of the symbol, because multiplying by a negative reflects every number across zero and reverses their order. So $-2x < 8$ becomes $\boxed{x > -4}$. Testing a value confirms it: $x = 0$ gives $0 < 8$, true, and 0 is in the solution set; $x = -10$ gives $20 < 8$, false, and -10 is not.

6. Solve $4(x - 2) = 3x + 5$ with reasons.

Solution: Distributive Property on the left: $4x - 8 = 3x + 5$. Subtract $3x$ from both sides — Subtraction Property of Equality (a variable term is a quantity like any other): $x - 8 = 5$. Add 8 to both sides — Addition Property of Equality: $\boxed{x = 13}$. Every line kept the same solution, so the last line's solution is the first line's solution.

7. $x^2 = 4x$, divided by x .

Solution: The Division Property of Equality requires the divisor to be *nonzero*, and here the divisor is x itself, which may be zero. Dividing by it silently assumes $x \neq 0$ and throws that possibility away, so the step is not legal as written; it loses a solution rather than gaining a false one.

Legal route: subtract $4x$ from both sides to get $x^2 - 4x = 0$, factor to $x(x - 4) = 0$, and apply the zero-product property: $\boxed{x = 0 \text{ or } x = 4}$. The discarded root really was a solution, since $0^2 = 4 \cdot 0$.

8. Ravi: $2x + 6 = 10 \rightarrow x + 6 = 5$.

Solution: Dividing both sides by 2 would be legal, but Ravi divided only *some* terms: the $2x$ and the 10 were halved while the 6 was left alone. A division property applies to each side as a whole, not term by term.

Ravi's equation solves to $x = -1$. Substituting that into the original left side gives $2(-1) + 6 = \boxed{4}$, and the original right side is 10, so $x = -1$ is not a solution of the original equation. The two equations are therefore not equivalent. Substituting into Ravi's own equation instead would give 5 and hide the error — always test a suspect step against the *original*.

9. Solve $-3(x - 2) \geq 9$.

Solution: Distributive Property: $-3x + 6 \geq 9$. Subtract 6 from both sides — Subtraction Property of Equality: $-3x \geq 3$. Divide both sides by -3 ; this is the line where the symbol turns around, because dividing by a negative reverses order. Result: $\boxed{x \leq -1}$. Check with $x = -5$: $-3(-5 - 2) = 21 \geq 9$, true.

10. One illegal line in the solution of $2(x + 3) = 4x - 8$.

Solution: The illegal line is the second, $2x + 3 = 4x - 8$. It should have used the Distributive Property, which multiplies the 2 by *both* terms inside the parentheses: $2(x + 3) = 2x + 6$, not $2x + 3$. The third line is then correct arithmetic applied to a wrong equation, which is why naming the first bad line matters.

Correct solution: $2x + 6 = 4x - 8$; subtract $2x$ from both sides: $6 = 2x - 8$; add 8 to both sides: $14 = 2x$; divide both sides by 2: $\boxed{x = 7}$.